HAEBOT / KIRO V2 power system comparison

Engineering review document, Rev A, 2026-11-13

Both designs are PROPOSED. Neither has been built or measured. KiCad ERC has not been run yet.

1. Project overview

- HAEBOT (also called KIRO) is a mobile dual-arm robot: Jetson Orin Nano Super, SO-101 arms, four mecanum wheels driven by ST3215-C018 bus servos through one Bus Servo Adapter board.
- V1 runs from AC mains through a 12 V SMPS, two fuse boxes and a 12 V to 5 V buck converter. It is built and is the reference for all load figures.
- V2 removes the mains cable. Two batteries are under consideration, so there are two independent designs.
- **V2-A** uses an Anker SOLIX C300 DC (A1726) power station with ready-made leads on its protected ports.
- **V2-B** uses a 12 V 45 Ah battery, keeps the V1 fuse-box layout and moves both arms to 12 V servos.

Contents

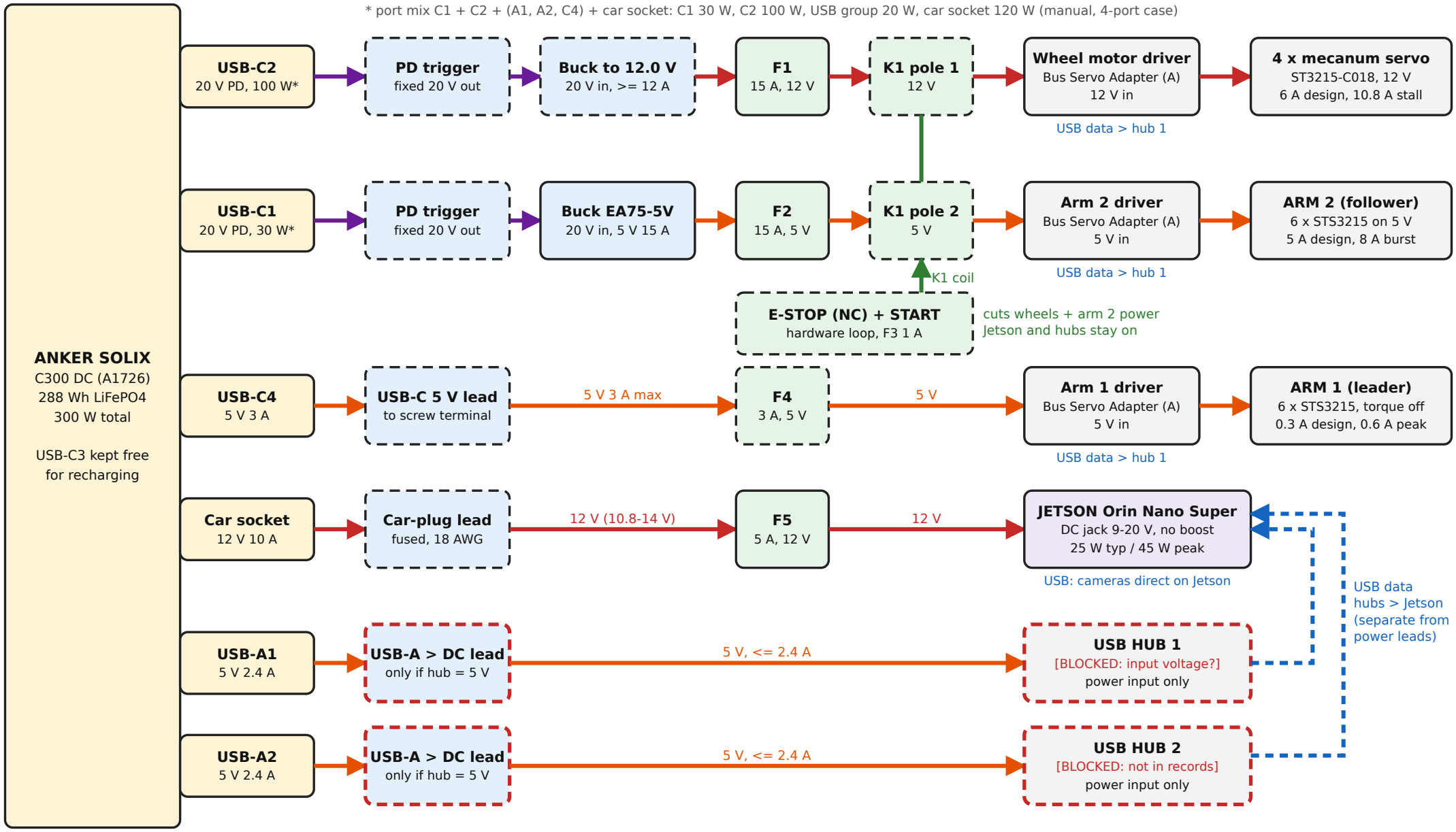
- Page 2: Anker block diagram. Page 3: Anker electrical schematic (A3).
- Page 4: EV battery block diagram. Page 5: EV battery electrical schematic (A3).
- Then: architecture comparison, power budget comparison, component and connection summary, electrical warnings, unresolved hardware items.

How to read the status words

Word	Meaning
Verified	Supported by a datasheet, a manual or a measurement recorded in the repository
Calculated	Derived from verified values with stated assumptions
Proposed	Not bought, not built, not tested
Blocked	Cannot be finalised until a hardware fact is supplied

HAEBOT / KIRO V2-A Anker SOLIX C300 DC power block diagram

Rev A, 2026-11-13. PROPOSED architecture, not bench-verified. Port limits from the Anker A1726 user manual; loads from V1 Rev F records.



LEGEND — 20 V USB-C PD — 12 V — 5 V — USB data — protection / E-stop control

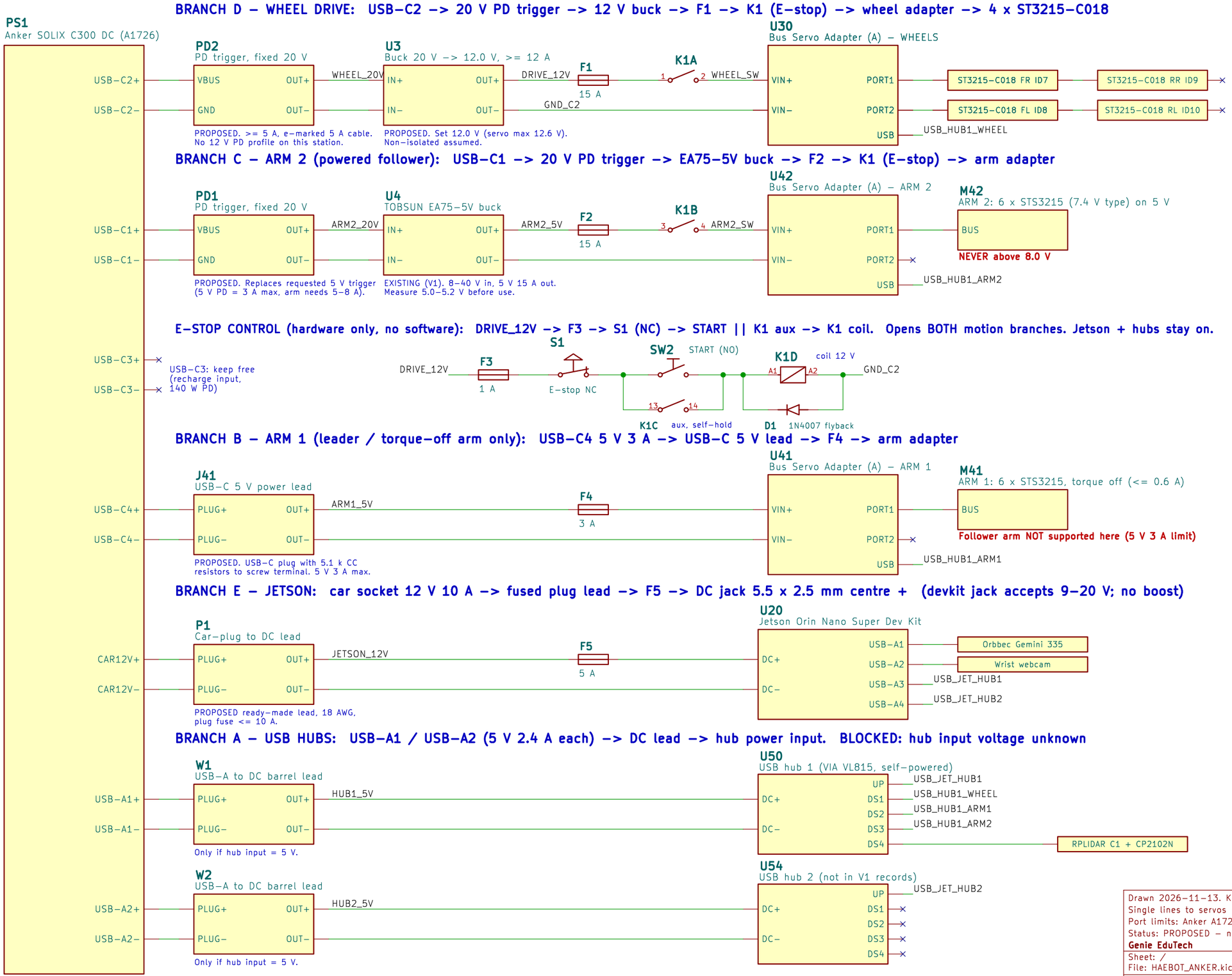
Borders: solid = existing V1 hardware dashed = PROPOSED (not bought / tested) red dashed = BLOCKED (needs a hardware fact)

Not supported as requested: 12 V PD trigger (station has no 12 V USB-C profile) and 5 V PD trigger for a follower arm (5 V PD is 3 A max).

HAEBOT / KIRO V2-A ANKER SOLIX C300 DC POWER SYSTEM

Electrical schematic, Rev A, 2026-11-13. STATUS: PROPOSED - build only after the bench checks in docs/verification_report.md

BRANCH D - WHEEL DRIVE: USB-C2 -> 20 V PD trigger -> 12 V buck -> F1 -> K1 (E-stop) -> wheel adapter -> 4 x ST3215-C018



BRANCH C - ARM 2 (powered follower): USB-C1 -> 20 V PD trigger -> EA75-5V buck -> F2 -> K1 (E-stop) -> arm adapter

E-STOP CONTROL (hardware only, no software): DRIVE_12V -> F3 -> S1 (NC) -> START || K1 aux -> K1 coil. Opens BOTH motion branches. Jetson + hubs stay on.

BRANCH B - ARM 1 (leader / torque-off arm only): USB-C4 5 V 3 A -> USB-C 5 V lead -> F4 -> arm adapter

BRANCH E - JETSON: car socket 12 V 10 A -> fused plug lead -> F5 -> DC jack 5.5 x 2.5 mm centre + (devkit jack accepts 9-20 V; no boost)

BRANCH A - USB HUBS: USB-A1 / USB-A2 (5 V 2.4 A each) -> DC lead -> hub power input. BLOCKED: hub input voltage unknown

ANKER PORT LIMITS (manual A1726)

USB-C1: 5/9/15 V 3 A, 20 V 5 A (100 W)
USB-C2/C3: + 28 V 5 A (140 W)
USB-C4: 5 V 3 A (15 W)
USB-A1/A2: 5 V 2.4 A (12 W each)
USB-A1 + A2 + C4 together: 20 W
Car socket: 12 V 10 A (120 W)
Total output: 300 W
NO 12 V USB-C profile, no PPS listed

PORT MIX USED ON THIS SHEET

C1 + C2 + (A1,A2,C4) + car socket
= manual 4-port case:
C1 30 W | C2 100 W | USB group 20 W
| car 120 W

CALCULATED MARGINS

Wheels: 78 W design < 100 W OK
Wheels: 141 W all-stall > 100 W
Arm 2: 28 W design < 30 W marginal
Arm 2: 44 W burst > 30 W
Jetson: 45 W peak < 120 W OK
Hubs + Arm 1: must stay < 20 W

WARNINGS

1. Never join the + of two ports.
2. Plugging or unplugging any port can renegotiate PD and drop the other outputs for a moment.
3. Set and measure U3 (12.0 V) and U4 (5.0-5.2 V) before any servo is connected.
4. Negatives of all ports become common through USB shields. Motor current must return on its own pair, not through USB.
5. Followers drop when K1 opens.

STATUS WORDS

EXISTING = part on the V1 robot
PROPOSED = not bought, not tested
BLOCKED = needs a hardware fact

Drawn 2026-11-13. KiCad ERC must be run in KiCad 8+ before release
Single lines to servos = 3-wire Molex 5264 bus cable (GND / VCC / SIG)
Port limits: Anker A1726 user manual. Loads: V1 Rev F records (kiro-wiring/docs)
Status: PROPOSED - not bench-verified. See docs/verification_report.md
Genie EduTech

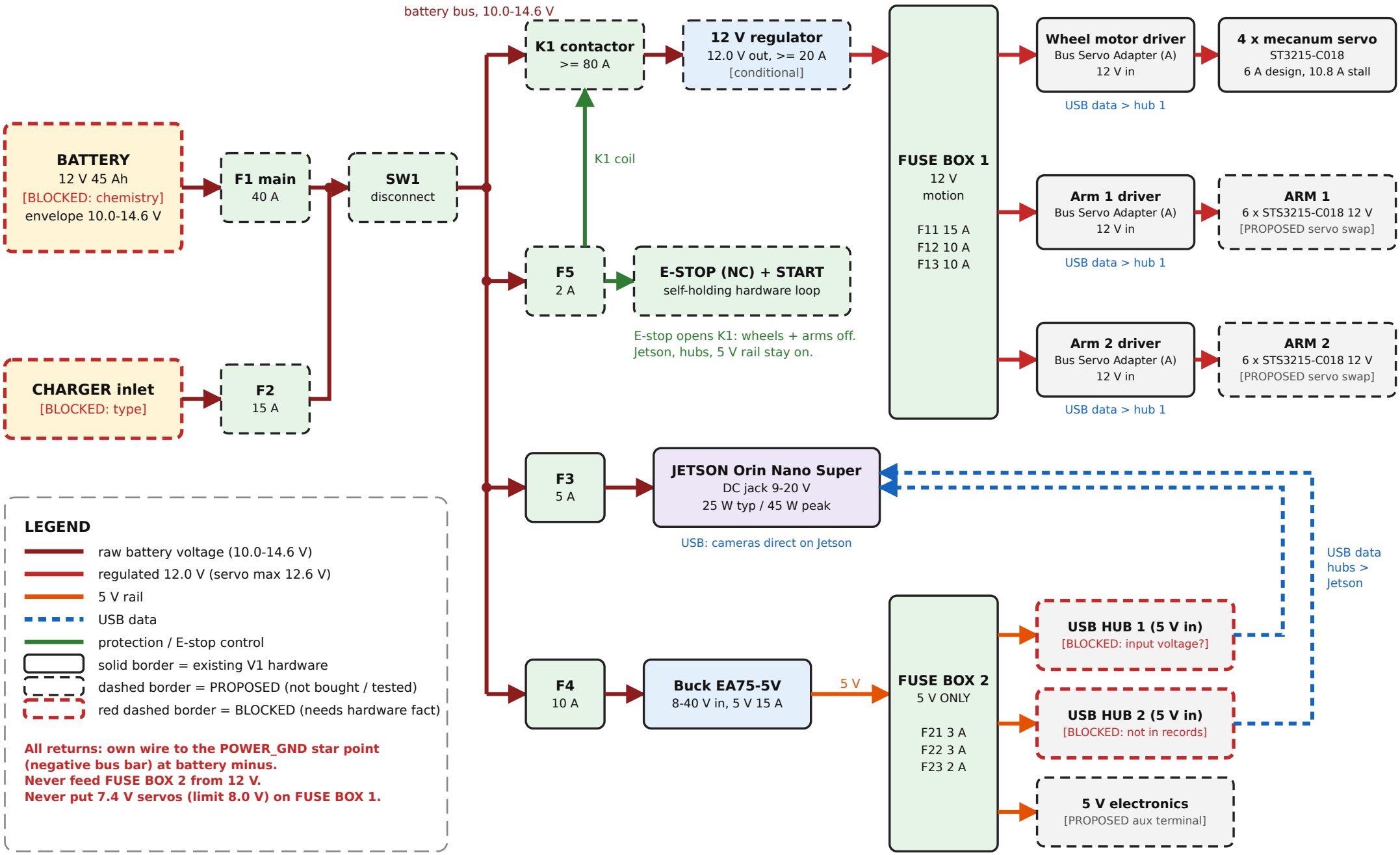
Sheet: /
File: HAEBOT_ANKER.kicad_sch

Title: HAEBOT / KIRO V2-A - Anker SOLIX C300 DC power system

Size: A3	Date: 2026-11-13	**Rev: A**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1

HAEBOT / KIRO V2-B 12 V 45 Ah EV battery power block diagram

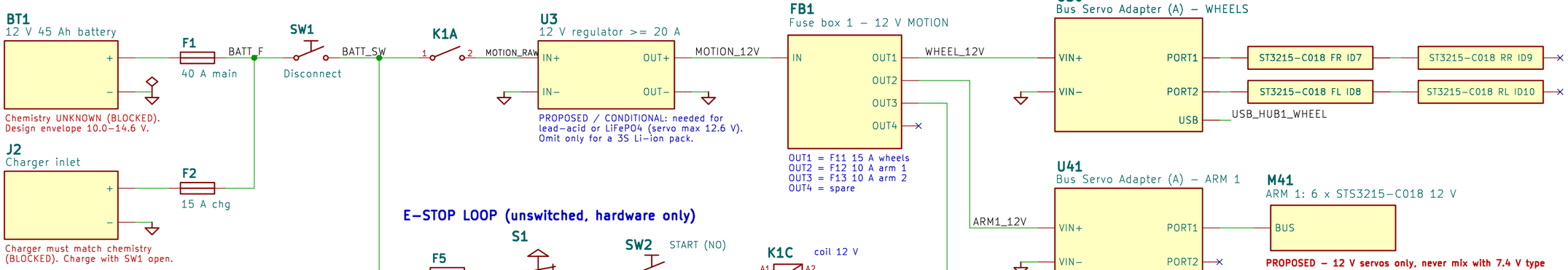
Rev A, 2026-11-13. PROPOSED architecture. Battery chemistry, BMS limits and charger are BLOCKED items; fuse and wire values are preliminary.



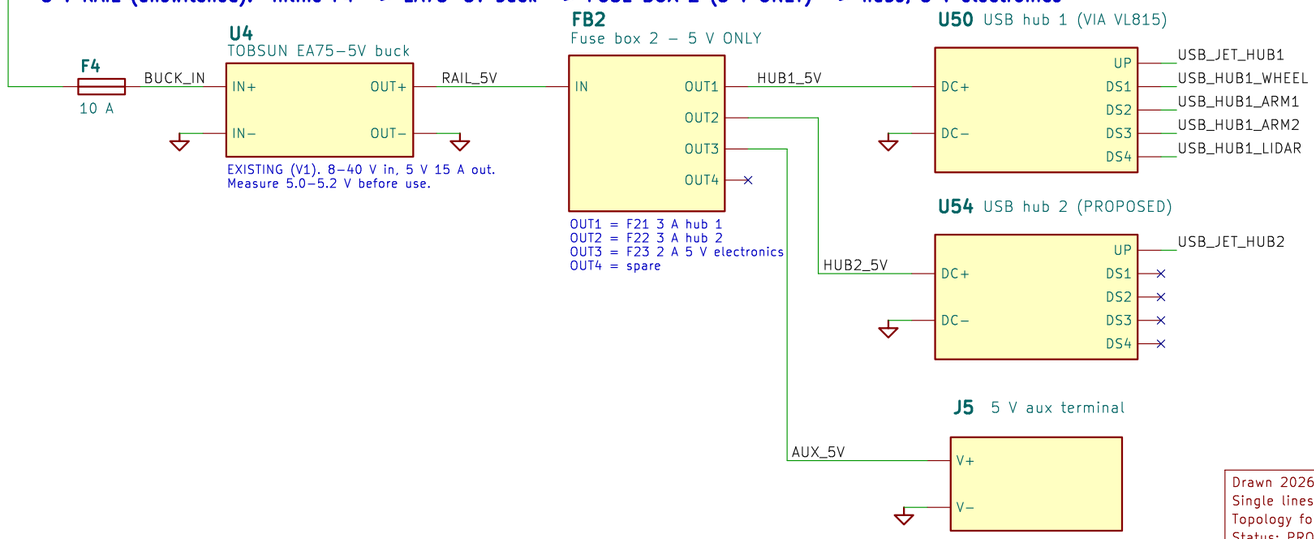
HAEBOT / KIRO V2-B 12 V 45 Ah EV BATTERY POWER SYSTEM

Electrical schematic, Rev A, 2026-11-13. STATUS: PROPOSED - battery chemistry, BMS limits and charger are BLOCKED items

MOTION POWER (cut by E-stop): battery -> F1 -> SW1 -> K1 -> 12 V regulator -> FUSE BOX 1 -> wheels, arm 1, arm 2



5 V RAIL (unswitched): inline F4 -> EA75-5V buck -> FUSE BOX 2 (5 V ONLY) -> hubs, 5 V electronics



SYMBOLS AND STATUS

Arrow-down symbol = POWER_GND:
own return wire to the star point
(negative bus bar) at battery -.
Never return motor current
through a USB cable.

EXISTING = on the V1 robot
PROPOSED = not bought / tested
BLOCKED = needs a hardware fact

E-STOP BEHAVIOUR

S1 pressed or wire broken:
K1 opens, wheels + both arms
lose power (arms drop).
Jetson, hubs, 5 V stay on.
Release S1: no restart until
START is pressed.

BEFORE BUILDING

Read battery + charger labels.
Fix F1, F2, K1, U3 after that.

On hub 1, port DS4:
USB_HUB1_LIDAR -> RPLIDAR C1 + CP2102N

DESIGN VALUES

Jetson 12 V: 25 / 36 / 45 W
Wheels: 1.2 / 6.0 / 10.8 A
Arm (12 V, each): 3 A design,
8 A peak (ESTIMATE)
5 V rail: about 10 W (assumed)
Battery: see docs/power_budget.md

WIRE (from V1 Rev F rules)

Battery to FB1: 8 AWG
Wheels, arms, buck: 14 AWG
Jetson: 18 AWG
Control loop: 20 AWG

WARNINGS

1. Never feed FB2 from 12 V.
2. Never connect 7.4 V servos
(limit 8.0 V) to FB1.
3. ST3215 max 12.6 V: battery
may reach 14.6 V -> U3.

Drawn 2026-11-13. KiCad ERC must be run in KiCad 8+ before release
Single lines to servos = 3-wire Molex 5264 bus cable (GND / VCC / SIG)
Topology follows V1 Rev F (two fuse boxes, EA75-5V buck); SMPS replaced by battery
Status: PROPOSED - battery chemistry, BMS and charger are not verified (BLOCKED items)
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Sheet: /
File: HAEBOT_EV.kicad_sch

Title: HAEBOT / KIRO V2-B - 12 V 45 Ah EV battery power system

Size: A3	Date: 2026-11-13	**Rev: A**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1

2. Architecture comparison

	V2-A Anker SOLIX C300 DC	V2-B 12 V 45 Ah EV battery
Energy source	288 Wh LiFePO4 station, 300 W total DC (verified, manual)	12 V 45 Ah, 540-576 Wh nominal; chemistry not identified (blocked)
Distribution	five station ports, one lead pair per branch	main fuse, disconnect, battery bus, two fuse boxes
Wheel supply	USB-C2 20 V PD, PD trigger, 12.0 V buck, F1 15 A	K1, 12.0 V regulator (conditional), fuse box 1 F11 15 A
Arm supply	existing 5 V servos. Leader on USB-C4 (5 V 3 A). Follower on USB-C1 through 20 V trigger and EA75-5V buck	new 12 V servos STS3215-C018 on fuse box 1, F12 and F13 10 A
Jetson supply	car socket 12 V 10 A, direct to the 9-20 V DC jack, F5 5 A	battery bus, direct to the 9-20 V DC jack, F3 5 A
5 V rail	EA75-5V for the follower arm only	EA75-5V, fuse box 2: hubs and 5 V electronics
USB hubs	USB-A ports (blocked: hub input voltage unknown)	fuse box 2 (blocked: hub input voltage unknown)
Emergency stop	2-pole contactor opens wheel and follower branches; hardware loop	battery contactor opens fuse box 1; hardware loop
Over-current protection	station port protection plus branch fuses	40 A main fuse plus branch fuses (values preliminary)
Charging	USB-C2 / USB-C3, 140 W PD each	blocked: charger not identified
New parts to buy	2 PD triggers, 12 V buck, contactor, E-stop, START, leads, holders	main fuse, disconnect, contactor, regulator, bus bar, E-stop, START, 12 servos
Changes to the robot	none to arms or software	12 servos replaced, both arms recalibrated
Main limit	port sharing: 100 W for wheels and 30 W for the follower in this port mix	battery data missing; servo rail must stay at or below 12.6 V

3. Power budget comparison

Loads are the V1 Rev F values. Anker usable energy: 288 Wh x 0.85 = 245 Wh (assumption). EV usable energy: Lead-acid (AGM / gel) 270 Wh, LiFePO4 4S 461 Wh, Li-ion 3S 400 Wh (assumed usable shares 50 / 80 / 80 %).

Scenario	Anker: power	Anker: runtime	EV: power	EV: Lead-acid	EV: LiFePO4 4S	EV: Li-ion 3S
1. Idle robot	38 W	6.4 h	36 W	7.5 h	12.8 h	11.1 h
2. Jetson and electronics active	50 W	4.9 h	64 W	4.2 h	7.3 h	6.3 h
3. Two arms operating	81 W	3.0 h	134 W	2.0 h	3.4 h	3.0 h
4. Robot driving, arms holding	137 W	1.8 h	147 W	1.8 h	3.1 h	2.7 h
5. Typical combined operation	63 W	3.9 h	73 W	3.7 h	6.3 h	5.4 h
6. High load	157 W	1.6 h	281 W	1.0 h	1.6 h	1.4 h

- The scenario definitions differ slightly between the designs (the EV design has two powered 12 V arms). Details are in each folder's docs/power_budget.md.
- Anker wheel branch: 78 W at design load against a 100 W port limit; 141 W with all four wheels stalled is not supported.
- Anker follower branch: 28 W at design load against 30 W; the 44 W burst is not supported.
- EV battery current: 23 A in the high-load case; 34 A at 12.0 V if everything peaks together.

4. Component and connection summary

Function	V2-A Anker	V2-B EV battery	Connection
Computer	Jetson Orin Nano Super Dev Kit (existing)	same	5.5 x 2.5 mm barrel, centre +
Wheel motor driver	Bus Servo Adapter (A), CH343 (existing, one board)	same	screw terminal, USB-C data
Wheel motors	4 x ST3215-C018 (existing)	same	Molex 5264, two chains
Arm drivers	2 x Bus Servo Adapter (A) on 5 V (existing)	same boards moved to 12 V	screw terminal, USB-C data
Arm servos	STS3215 7.4 V type on 5 V (existing)	12 x STS3215-C018 12 V (proposed)	Molex 5264
5 V converter	TOBSUN EA75-5V, fed with 20 V (existing)	TOBSUN EA75-5V, fed from the battery (existing)	flying leads
12 V regulation	buck 20 V to 12.0 V, 12 A or more (proposed)	buck-boost 12.0 V, 20 A or more (proposed, conditional)	screw terminals
Fuse boxes	inline holders only	2 x Dorhea 4-way (existing)	blade fuses
E-stop	mushroom NC, START, 2-pole contactor (proposed)	mushroom NC, START, battery contactor (proposed)	screw terminals
USB hubs	hub 1 VIA VL815 (existing), hub 2 (proposed)	same	data to Jetson; power input blocked

5. Important electrical warnings

- Follower servos of the 7.4 V type trip above 8.0 V. Never connect them to a 12 V rail.
- ST3215-C018 servos accept at most 12.6 V. A lead-acid or LiFePO4 battery exceeds this; use the 12.0 V regulator.
- The Anker station has no 12 V USB-C profile. A 12 V PD trigger will not give 12 V.
- USB-C delivers at most 3 A at 5 V. A powered arm cannot run from a 5 V USB-C source.
- Never join the positive outputs of two station ports. Never feed fuse box 2 from 12 V.
- Motor current must return on its own wire, never through a USB cable.
- The E-stop is a hardware loop. When it opens, the powered arms drop under gravity.
- A 45 Ah battery can deliver hundreds of amperes into a short. Fit the main fuse within 150 mm of the positive terminal.

6. Unresolved hardware verification items

#	Item	Affects	What settles it
1	EV battery chemistry, voltage range, BMS limits, connector	V2-B: main fuse, regulator, cut-off	photo of the battery label
2	Charger type and connector	V2-B: charging connection, F2	photo of the charger label
3	USB hub input voltage, current, plug	both: hub power branch	photo of hub and adapter labels
4	Second USB hub: does it exist?	both	your answer
5	Which arms are arm 1 and arm 2 (leader / follower)	V2-A port assignment; V2-B servo count	your answer
6	Trial fit and current of one STS3215-C018 in an arm joint	V2-B: brackets, F12, F13	bench test
7	20 V available on USB-C1 when capped at 30 W; behaviour on hot-plug	V2-A: follower branch	bench test with a USB-C meter
8	Station port negatives common internally?	V2-A: grounding	continuity measurement
9	Part numbers: PD triggers, bucks, contactors, E-stop, disconnect	both BOMs	selection and purchase
10	Jetson carrier is the stock P3768 board	both: 9-20 V jack range	visual check
11	KiCad ERC in KiCad 8 or newer	both schematics	run ERC, review every message
12	Recommended Anker alternative: arm buck on the car socket	V2-A: 140 W for the wheels	your decision